

LICHTENBURG, South Africa

WMO: 683410

Lat: 26.133S

Lon: 26.164E

Elev: 4872

StdP: 12.29

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.3	34.6	8.7	10.3	54.8	11.8	12.0	54.2	19.9	62.2	18.1	63.9	3.7	40	0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	21.1	91.8	61.5	89.1	61.2	86.7	60.7	67.9	79.5	67.0	78.1	66.2	76.8	9.3	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	110.1	71.5	63.9	106.8	70.8	63.1	103.9	70.2	35.4	79.8	34.6	77.9	33.9	76.4	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.3	14.4	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	23.0	69.4	4.6	0.9	19.7	70.1	17.0	70.6	14.4	71.2	11.1	71.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.4	72.2	71.4	68.7	63.5	57.5	52.9	52.3	57.9	64.7	69.0	70.4	72.6		
	DBStd	8.72	3.80	3.59	3.68	4.54	4.46	4.52	4.77	6.16	6.16	5.87	5.80	4.55		
	HDD50	71	0	0	0	0	6	22	32	10	1	0	0	0		
	HDD65	1441	1	1	11	79	233	363	393	228	79	30	20	3		
	CDD50	5322	687	598	579	406	239	109	103	256	442	589	613	701		
	CDD65	1220	223	180	125	35	1	0	0	9	70	155	184	238		
	CDH74	9772	1570	1113	753	247	65	5	3	175	776	1538	1574	1953		
	CDH80	3119	532	315	145	32	1	0	0	11	208	579	546	750		
(14) Wind	WSAvg	6.6	6.6	6.0	5.6	5.4	5.0	5.6	5.9	7.4	7.8	7.9	8.1	7.7		
(15) Precipitation	PrecAvg	21.5	4.2	3.3	3.2	2.2	0.5	0.2	0.0	0.2	0.6	1.7	3.4	3.2		
	PrecMax	40.1	10.2	9.0	8.5	6.5	2.0	1.7	0.6	3.2	3.0	5.9	8.6	8.2		
	PrecMin	9.1	0.6	0.0	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.5		
	PrecStd	6.7	2.2	2.1	2.0	1.7	0.7	0.4	0.1	0.6	0.7	1.3	2.0	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.3	90.9	87.3	83.7	79.1	73.8	73.6	81.4	88.4	93.8	93.4	96.0		
		MCWB	64.3	62.8	62.5	60.5	54.5	51.1	49.5	54.9	56.6	60.2	60.6	62.2		
	2%	DB	89.9	87.5	84.0	80.1	76.0	70.4	70.0	78.5	85.2	89.6	89.6	90.9		
		MCWB	63.5	63.2	62.6	60.0	54.0	49.7	48.6	53.7	55.7	58.7	60.3	62.8		
	5%	DB	86.2	84.2	81.0	77.1	73.2	67.7	67.1	75.3	82.3	86.7	86.4	87.4		
		MCWB	63.5	64.3	63.0	59.5	53.5	49.1	48.0	52.1	54.7	57.9	59.6	62.2		
	10%	DB	82.5	80.9	78.5	73.9	69.8	64.6	63.9	71.6	78.6	83.3	83.1	84.0		
		MCWB	64.2	64.8	62.5	58.7	52.5	47.9	46.8	50.0	53.5	56.9	59.8	62.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.3	69.3	67.5	65.2	58.9	53.6	52.7	57.3	62.0	63.7	66.5	67.5		
		MCDB	82.7	81.3	78.9	76.1	73.1	67.8	67.2	76.5	75.1	83.5	80.7	84.0		
	2%	WB	68.0	68.1	66.4	63.6	56.8	51.9	50.9	55.6	59.0	61.7	64.9	66.3		
		MCDB	79.2	78.7	77.8	74.1	69.9	66.4	65.5	73.8	77.3	79.8	79.2	81.1		
	5%	WB	67.0	67.1	65.3	62.3	55.4	50.5	49.4	53.8	57.1	60.4	63.7	65.3		
		MCDB	77.4	76.9	75.6	72.2	68.9	64.4	63.8	69.5	76.1	78.2	77.8	79.3		
	10%	WB	66.1	66.2	64.2	60.8	53.9	48.8	48.0	52.1	55.5	59.4	62.4	64.5		
		MCDB	75.9	75.3	73.1	70.1	66.8	61.9	62.1	67.1	73.3	76.5	76.0	77.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.3	18.4	19.5	21.1	26.7	27.7	26.7	27.3	27.1	25.0	23.0	21.1		
		MCDBR	26.1	25.2	24.1	25.3	30.3	30.1	29.0	29.9	30.4	29.3	27.4	27.0		
		MCWBR	8.5	8.8	8.9	9.4	13.1	14.1	13.6	13.6	12.7	11.3	9.4	9.0		
	5% WB	MCDBR	17.6	17.8	18.9	19.6	25.7	26.8	25.5	26.3	25.6	23.5	21.8	20.5		
		MCWBR	6.8	7.3	8.0	8.6	11.8	13.4	12.8	12.7	11.1	9.6	8.0	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.362	0.367	0.352	0.316	0.279	0.264	0.263	0.316	0.360	0.386	0.340	0.348		
	taud		2.397	2.378	2.405	2.475	2.553	2.565	2.541	2.379	2.263	2.229	2.414	2.425		
	Ebn at Noon		311	304	299	297	296	295	299	292	292	295	316	316		
	Edh at Noon		41	40	37	32	26	25	26	34	42	46	40	40		
(44) All-Sky Solar Radiation	RadAvg		2198	2033	1817	1537	1395	1256	1368	1629	1969	2141	2263	2291		
	RadStd		224	188	155	130	61	56	36	77	79	129	180	138		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page